

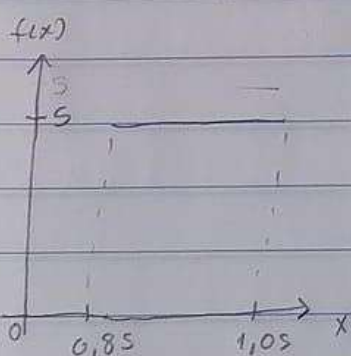
ATIVIDADE 6

Pág. 97 - EXER. 4.7 (todas as letras)

$$4.7) a) \int_{0,85}^{1,05} k dx = 1; \quad k(1,05 - 0,85) = 1; \quad k \cdot 0,2 = 1$$

$$k = \frac{1}{0,2} = 5 //$$

$$f(x) = \begin{cases} 5, & 0,85 \leq x \leq 1,05 \\ 0, & \text{caso contrário} \end{cases}$$



$$k = \frac{1}{1,05 - 0,85} = \frac{1}{0,2} = 5$$

$$b) E(x) = \mu = \int_{0,85}^{1,05} x \cdot 5 dx = \frac{5x^2}{2} \Big|_{0,85}^{1,05} = \frac{5}{2} [(1,05)^2 - (0,85)^2] =$$

$$= \frac{5}{2} [1,1025 - 0,7225] = \frac{5}{2} \cdot 0,38 = 0,95 //$$

$$E(x) = \int_{0,85}^{1,05} x^2 \cdot 5 dx = \frac{5}{3} x^3 \Big|_{0,85}^{1,05} = \frac{5}{3} [(1,05)^3 - (0,85)^3] =$$

$$= \frac{5}{3} [1,157625 - 0,614125] = 0,5435 \cdot \frac{5}{3} = 0,90583$$

$$VAR(x) = \sigma^2 = 0,90583 - [0,95]^2 = 0,0333$$

$$c) P(a \leq x \leq b) = \int_a^b f(x) dx$$

$$P(x < 1) = \int_{0,85}^1 5 dx = 5x \Big|_{0,85}^1 = 5(1 - 0,85) = 5 \cdot 0,15 = 0,75 //$$

$$P(x < 1) = 0,75$$

d) $P(X > 1) = 0,25$

$$P(X > 1) = \int_1^{1,05} S dx = Sx \Big|_1^{1,05} = 5(1,05 - 1) = 5 \cdot 0,05 = 0,25$$

$$E = 200 \cdot 0,25 = 50 //$$